

OFFLINE SYNTHESIS · training interventions + validation schedules

Actual rollout actions

Contacts and rewards fixed
Reached boundary $B = (F, X)$
Chosen action + challengers

Cancel, then certify

Common components cancel
Full residual: $\Delta \in [L\Delta, U\Delta]$
Strict sign / tie / unknown

Strict requirements

$h(\phi(x^+)) > h(\phi(x^-))$
Reversal + preservation
Replay all earlier requirements

Exact quotient diagnosis

Cycle $\rightarrow$ no scalar score of ϕ

$$\begin{aligned}x_1^+ \rightarrow x_1^- &\equiv \phi \quad x_2^+ \\x_2^+ \rightarrow x_2^- &\equiv \phi \quad x_1^+\end{aligned}$$

LLM: typed catalogue

NodeSet $\rightarrow$ EdgeSet $\rightarrow$ Number

Minimum-cost master

$$\min \sum c_f z_f; \sum d_{Wf} z_f \geq 1$$

Full quotient separation

Surviving cycle: add witness
DAG: observed repair passes
Budget / no feature: unresolved

Concrete arcs + equality joins

cycle: add cut; resolve master

DAG

Freeze (ϕ , h , kernel)

Score-sliced shared DAG

LLM rules: actual-action audit + joint selection

Reach / escape / pool regret; consistency $s(P)$, quality $q(P)$, work $c(P)$

ONLINE

Current graph $\rightarrow$ demanded inputs $\rightarrow$ frozen $h \rightarrow$ argmax $\rightarrow$ delete closed neighborhood